ArrayList and LinkedList:

1. Create an ArrayList to store the names of students in a class. Add, remove, and print the list of students.
    - Initialize an empty ArrayList to store examinee names.
    - Add the names of five examiners participating in the exam to the ArrayList.
    - Remove the name of the examinee who withdrew from the exam.
    - Print the updated list of participants.

```
import java.util.ArrayList;

public class ParticipantManager {
    public static void main(String[] args) {
        ArrayList<String> students = new ArrayList<>();
        students.add("Ram");
        students.add("Bahadur");
        students.add("Shyam");
        students.add("Pooja");
        students.add("Nani maiya");
        students.remove("Shyam Bahadur");

        ArrayList<String> examinees = new ArrayList<>();
        examinees.add("Ritesh");
        examinees.add("Shaurabh");
        examinees.add("Aayush");
        examinees.add("Pranish");
        examinees.add("Yajju");
        examinees.remove("Nissan");
```

```java
        System.out.println("Students:");
        for(String student : students) {
            System.out.println(student);
        }

        System.out.println("Examinees:");
        for(String name : examinees) {
            System.out.println(name);
        }
    }
}
```

```
PS E:\jaca\JavaTut6>
PS E:\jaca\JavaTut6>  e:; cd 'e:\jaca\JavaTut6'; & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\wo
rkspaceStorage\af8e616639675acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'ParticipantManager'
Students:
Ram
Bahadur
Shyam
Pooja
Nani maiya
Examinees:
Ritesh
Shaurabh
Aayush
Pranish
Yajju
PS E:\jaca\JavaTut6>
```

2. Write a program to insert elements into the linked list at the first and last positions. Also check if the linked list is empty or not.

```java
import java.util.LinkedList;

public class LinkedListDemo {
    public static void main(String[] args) {
        LinkedList<String> list = new LinkedList<>();

        System.out.println("Is list empty? " + list.isEmpty());
```

```java
        list.addFirst("FirstElement");
        list.addLast("LastElement");

        System.out.println("Is list empty? " + list.isEmpty());

        for(String item : list) {
            System.out.println(item);
        }
    }
}
```

```
Cannot load PSReadline module.  Console is running without PSReadline.
PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'LinkedListDemo'
Is list empty? true
Is list empty? false
FirstElement
LastElement
PS E:\jaca\JavaTut6>
```

3. Rotate the elements of an ArrayList to the right by a given number of positions. For example, if the ArrayList is [1, 2, 3, 4, 5] and you rotate it by 2 positions, the result should be [4, 5, 1, 2, 3].

```java
import java.util.ArrayList;
import java.util.Collections;

public class ArrayListRotator {
    public static void main(String[] args) {
```

```java
        ArrayList<Integer> list = new ArrayList<>();
        Collections.addAll(list, 1, 2, 3, 4, 5);

        int rotateBy = 2;
        Collections.rotate(list, rotateBy);

        for (int num : list) {
            System.out.print(num + " ");
        }
    }
}
```

```
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PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'ArrayListRotator'
4 5 1 2 3
PS E:\jaca\JavaTut6>
```

4. Write a program to declare a linkedList, colors to store String. Insert five colors into the linked list.
    a. Iterate and print all the colors.
    b. Check if “Red” exists in the linkedList or not.
    c. Shuffle the elements of the list and print them.
    d. Print the LinkedList in ascending order

```
import java.util.Collections;

import java.util.LinkedList;


public class ColorLinkedList {

    public static void main(String[] args) {

        LinkedList<String> colors = new LinkedList<>();

        colors.add("Blue");

        colors.add("Green");

        colors.add("Yellow");

        colors.add("Red");

        colors.add("Orange");


        for (String color : colors) {

            System.out.println(color);

        }
```

```
        System.out.println("Contains Red? " + colors.contains("Red"));


        Collections.shuffle(colors);

        System.out.println("Shuffled:");

        for (String color : colors) {

            System.out.println(color);

        }


        Collections.sort(colors);

        System.out.println("Sorted:");

        for (String color : colors) {

            System.out.println(color);

        }

    }

}
```

```
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PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'ColorLinkedList'
Blue
Green
Yellow
Red
Orange
Contains Red? true
Shuffled:
Orange
Blue
Red
Yellow
Green
Sorted:
Blue
Green
Orange
Red
Yellow
PS E:\jaca\JavaTut6>
```

Stack:

5. Create a Stack to manage a sequence of tasks. Implement the following operations:
    a. Push the tasks "Read", "Write", and "Code" onto the stack.
    b. Pop a task from the stack.
    c. Push tasks "Debug" and "Test" onto the stack.
    d. Peek at the top task without removing it.
    e. Print the stack.

```java
import java.util.Stack;

public class TaskManager {
    public static void main(String[] args) {
        Stack<String> tasks = new Stack<>();
```

```java
        tasks.push("Read");
        tasks.push("Write");
        tasks.push("Code");

        tasks.pop();

        tasks.push("Debug");
        tasks.push("Test");

        System.out.println("Top task: " + tasks.peek());

        System.out.println("Tasks stack:");
        for (String task : tasks) {
            System.out.println(task);
        }
    }
}
```

```
Cannot load PSReadline module.  Console is running without PSReadline.
PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'TaskManager'
Top task: Test
Tasks stack:
Read
Write
Debug
Test
PS E:\jaca\JavaTut6>
```

6. Write a program that reverses the order of words in a sentence using a Stack. For example, if the input is "Hello World", the output should be "World Hello".

```java
import java.util.Stack;
```

```
public class ReverseWords {

    public static void main(String[] args) {

        String sentence = "Hello World";

        String[] words = sentence.split(" ");


        Stack<String> stack = new Stack<>();

        for (String word : words) {

            stack.push(word);

        }


        StringBuilder reversed = new StringBuilder();

        while (!stack.isEmpty()) {

            reversed.append(stack.pop());

            if (!stack.isEmpty()) {

                reversed.append(" ");

            }

        }


        System.out.println(reversed.toString());

    }

}
```

```
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PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'ReverseWords'
World Hello
PS E:\jaca\JavaTut6>
```

Queue

7. Imagine a scenario where a printer is managing print jobs. Create a Queue to handle these print jobs. Implement the following operations:
    - Enqueue print jobs "Document1", "Document2", and "Document3" into the print queue.
    - Dequeue a print job from the front of the queue.
    - Enqueue print jobs "Document4" and "Document5" into the print queue.
    - Peek at the next print job without removing it.
    - Print the list of print jobs in the queue.

```
import java.util.LinkedList;
```

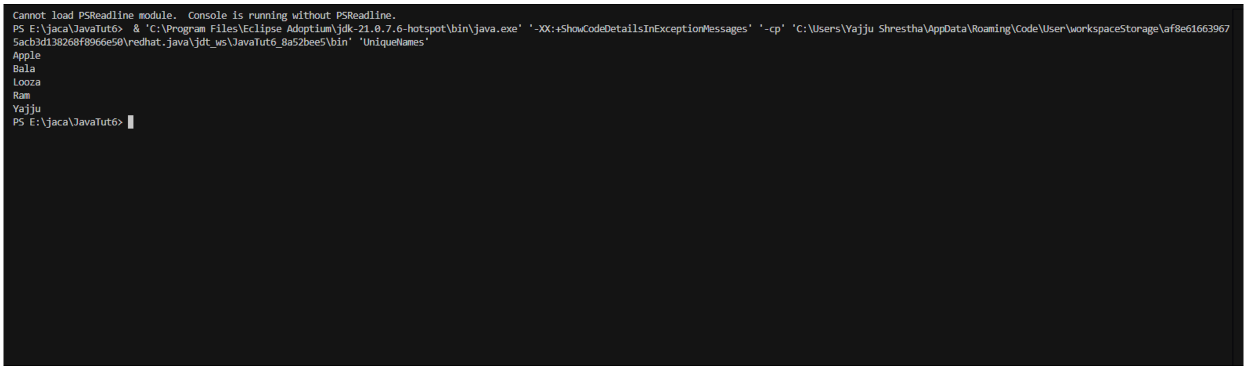
```java
import java.util.Queue;


public class PrintQueueManager {

    public static void main(String[] args) {

        Queue<String> printQueue = new LinkedList<>();


        printQueue.add("Document1");

        printQueue.add("Document2");

        printQueue.add("Document3");


        printQueue.poll();


        printQueue.add("Document4");

        printQueue.add("Document5");


        System.out.println("Next print job: " + printQueue.peek());


        System.out.println("Print queue:");

        for (String job : printQueue) {

            System.out.println(job);

        }

    }

}
```

```
Cannot load PSReadline module.  Console is running without PSReadline.
PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'PrintQueueManager'
Next print job: Document2
Print queue:
Document2
Document3
Document4
Document5
PS E:\jaca\JavaTut6>
```

Set Operations

8. Implement a TreeSet to store unique names in alphabetical order.

```
import java.util.TreeSet;


public class UniqueNames {

    public static void main(String[] args) {

        TreeSet<String> names = new TreeSet<>();

        names.add("Ram");

        names.add("Bala");

        names.add("Yajju");

        names.add("Looza");

        names.add("Apple");

        for (String name : names) {

            System.out.println(name);

        }

    }

}
```



9. Consider a scenario where you have two sets, each representing a group of animals. Implement a Java program to perform set operations (Union, Intersection, and Difference) on these sets:
   - Initialize two HashSet objects: **set1** with elements "Dog," "Cat," "Elephant," and "Lion," and **set2** with elements "Cat," "Giraffe," "Dog," and "Monkey."
   - Implement a method performUnion that takes two sets and returns their union.
   - Implement a method performIntersection that takes two sets and returns their intersection.
   - Implement a method performDifference that takes two sets and returns the difference of the first set from the second set.
   - Print the original sets, the union, intersection, and difference of the sets.

```
import java.util.HashSet;

import java.util.Set;


public class AnimalSetOperations {

    public static Set<String> performUnion(Set<String> set1, Set<String> set2) {

        Set<String> union = new HashSet<>(set1);

        union.addAll(set2);

        return union;

    }


    public static Set<String> performIntersection(Set<String> set1, Set<String> set2) {

        Set<String> intersection = new HashSet<>(set1);

        intersection.retainAll(set2);

        return intersection;

    }


    public static Set<String> performDifference(Set<String> set1, Set<String> set2) {

        Set<String> difference = new HashSet<>(set1);

        difference.removeAll(set2);

        return difference;
```

```
    }


    public static void main(String[] args) {

        Set<String> set1 = new HashSet<>();

        Set<String> set2 = new HashSet<>();


        set1.add("Dog");

        set1.add("Cat");

        set1.add("Elephant");

        set1.add("Lion");


        set2.add("Cat");

        set2.add("Giraffe");

        set2.add("Dog");

        set2.add("Monkey");


        System.out.println("Set1: " + set1);

        System.out.println("Set2: " + set2);


        System.out.println("Union: " + performUnion(set1, set2));

          System.out.println("Intersection: " + performIntersection(set1,
set2));

                  System.out.println("Difference  (set1  -  set2):  "  +
performDifference(set1, set2));
```

```
        }


}
```

```
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PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'AnimalSetOperations'
Set1: [Cat, Elephant, Lion, Dog]
Set2: [Cat, Monkey, Dog, Giraffe]
Union: [Cat, Elephant, Monkey, Lion, Dog, Giraffe]
Intersection: [Cat, Dog]
Difference (set1 - set2): [Elephant, Lion]
PS E:\jaca\JavaTut6>
```
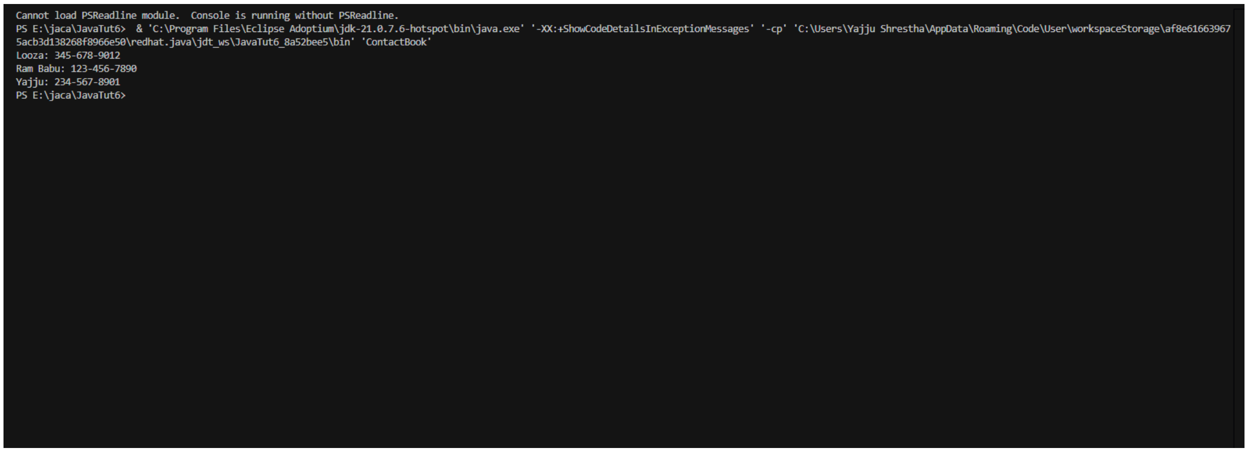

Map(HashMap, LinkedHashMap, TreeMap):

10. Write a program that uses a HashMap to store contact information (name and phone number).

```
import java.util.HashMap;

import java.util.Map;


public class ContactBook {

    public static void main(String[] args) {

        HashMap<String, String> contacts = new HashMap<>();


        contacts.put("Ram Babu", "123-456-7890");

        contacts.put("Yajju", "234-567-8901");

        contacts.put("Looza", "345-678-9012");


        for (Map.Entry<String, String> entry : contacts.entrySet()) {

            System.out.println(entry.getKey() + ": " + entry.getValue());

        }

    }

}
```



11. Imagine a scenario where you are managing information about countries and their capitals using a HashMap. Perform the following tasks:
    - Initialize a HashMap called countryCapitals to store the capitals of different countries. Add at least five country-capital pairs.
    - Implement a method called printMap that takes a HashMap and prints all the key-value pairs.
    - Implement a method called getCapital that takes a country name as a parameter and returns its capital from the countryCapitals map.
    - Implement a method called containsCapital that takes a capital name as a parameter and returns

whether that capital exists in the countryCapitals map.

- Iterate through the countryCapitals map and print each country and its capital.

```
import java.util.HashMap;

import java.util.Map;


public class CountryCapitalManager {

        private static HashMap<String, String> countryCapitals = new
HashMap<>();


    public static void printMap(HashMap<String, String> map) {

        for (Map.Entry<String, String> entry : map.entrySet()) {

                        System.out.println(entry.getKey() + " -> " +
entry.getValue());

        }

    }


    public static String getCapital(String country) {

        return countryCapitals.get(country);

    }


    public static boolean containsCapital(String capital) {
```

```
        return countryCapitals.containsValue(capital);
    }


    public static void main(String[] args) {
        countryCapitals.put("Nepal", "Kathmandu");
        countryCapitals.put("USA", "Washington D.C.");
        countryCapitals.put("France", "Paris");
        countryCapitals.put("Japan", "Tokyo");
        countryCapitals.put("India", "New Delhi");

        printMap(countryCapitals);


        System.out.println("Capital of Japan: " + getCapital("Japan"));
                    System.out.println("Contains capital Paris? " +
containsCapital("Paris"));


        for (Map.Entry<String, String> entry : countryCapitals.entrySet())
{
            System.out.println("Country: " + entry.getKey() + ", Capital:
" + entry.getValue());
        }
    }
}
```

Collection Algorithm

Sorting

12. Write a program that sorts an array of integers using the sort() method. Also try sorting in reverse order.

```
import java.util.Arrays;

import java.util.Collections;


public class ArraySorter {

    public static void main(String[] args) {

        Integer[] numbers = {5, 2, 8, 3, 1};


        Arrays.sort(numbers);
```

```java
        System.out.println("Sorted ascending:");

        for (int num : numbers) {

            System.out.print(num + " ");

        }

        System.out.println();



        Arrays.sort(numbers, Collections.reverseOrder());

        System.out.println("Sorted descending:");

        for (int num : numbers) {

            System.out.print(num + " ");

        }

    }

}
```

```
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5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'ArraySorter'
Sorted ascending:
1 2 3 5 8
Sorted descending:
8 5 3 2 1
PS E:\jaca\JavaTut6>
```

13. Write a program that sorts an array list of strings of colors using the sort() method. Also try sorting in reverse order.

```
import java.util.ArrayList;

import java.util.Collections;


public class ColorSorter {

    public static void main(String[] args) {

        ArrayList<String> colors = new ArrayList<>();

            Collections.addAll(colors, "Blue", "Red", "Green", "Yellow",
"Orange");


        Collections.sort(colors);

        System.out.println("Sorted ascending:");

        for (String color : colors) {

            System.out.print(color + " ");

        }

        System.out.println();


        Collections.sort(colors, Collections.reverseOrder());

        System.out.println("Sorted descending:");
```

```
        for (String color : colors) {

            System.out.print(color + " ");

        }

    }

}
```

```
Cannot load PSReadline module.  Console is running without PSReadline.
PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'ColorSorter'
Sorted ascending:
Blue Green Orange Red Yellow
Sorted descending:
Yellow Red Orange Green Blue
PS E:\jaca\JavaTut6>
```

Binary search

14. Write a program to initialize an ArrayList with a set of integers. Implement a binary search algorithm to find a particular integer.

```
import java.util.ArrayList;

import java.util.Collections;


public class BinarySearchExample {
```

```
    public static int binarySearch(ArrayList<Integer> list, int target) {

        int left = 0, right = list.size() - 1;

        while (left <= right) {

            int mid = left + (right - left) / 2;

            if (list.get(mid) == target) {

                return mid;

            } else if (list.get(mid) < target) {

                left = mid + 1;

            } else {

                right = mid - 1;

            }

        }

        return -1;

    }


    public static void main(String[] args) {

        ArrayList<Integer> numbers = new ArrayList<>();

        Collections.addAll(numbers, 10, 20, 30, 40, 50, 60);


        Collections.sort(numbers);


        int target = 40;
```

```
        int index = binarySearch(numbers, target);


        if (index != -1) {

              System.out.println("Found " + target + " at index " +
index);

        } else {

            System.out.println(target + " not found");

        }

    }

}
```

```
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PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'BinarySearchExample'
Found 40 at index 3
PS E:\jaca\JavaTut6>
```

Regular Expressions

15. Write a Java program to check whether a string contains only a certain set of characters (in this case a-z, A-Z and 0-9).

```java
public class CharacterCheck {
    public static boolean isValidString(String str) {
        return str.matches("[a-zA-Z0-9]+");
    }

    public static void main(String[] args) {
        String test1 = "Hello123";
        String test2 = "Hello_123";

        System.out.println(isValidString(test1));  // true
        System.out.println(isValidString(test2));  // false
    }
}
```

```
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PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'CharacterCheck'
true
false
PS E:\jaca\JavaTut6>
```

16. Write a Java program to find the sequence of one upper case letter followed by lower case letters. Z

```java
import java.util.regex.Matcher;
import java.util.regex.Pattern;

public class UpperLowerSequenceFinder {
    public static void main(String[] args) {
        String input = "Abc Def GHI Jklm Nop QrS";
```

```java
        Pattern pattern = Pattern.compile("[A-Z][a-z]+");
        Matcher matcher = pattern.matcher(input);

        while (matcher.find()) {
            System.out.println(matcher.group());
        }
    }
}
```

```
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PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'UpperLowerSequenceFinder'
Abc
Def
Jklm
Nop
Qr
PS E:\jaca\JavaTut6>
```

17. Develop a Java program to check if a given string represents a file with a ".txt" extension.

```java
public class TxtFileChecker {

    public static boolean isTxtFile(String filename) {

                              return    filename    !=    null    &&
filename.toLowerCase().endsWith(".txt");

    }



    public static void main(String[] args) {

        System.out.println(isTxtFile("document.txt"));  // true
```

```java
        System.out.println(isTxtFile("image.png"));      // false

        System.out.println(isTxtFile("notes.TXT"));      // true

    }

}
```

```
Cannot load PSReadline module.  Console is running without PSReadline.
PS E:\jaca\JavaTut6>  & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.7.6-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Yajju Shrestha\AppData\Roaming\Code\User\workspaceStorage\af8e61663967
5acb3d138268f8966e50\redhat.java\jdt_ws\JavaTut6_8a52bee5\bin' 'TxtFileChecker'
true
false
true
PS E:\jaca\JavaTut6>
```